The first coupling event fails with probability of order one over the logarithm

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doi:10.5802/jep.317; CC BY 4.0

Editorial note. The counted statement is the author's labelled lem environment at source0.tex lines 886-889; its proof environment opens at line 891, the line after the blank separator, so the pairing is by adjacency and the retained proof range 891-961 is the whole environment.

Editorial note. Four environments are retained as context because the counted proof depends on them: eq:mgf theta infty (source lines 728-731), the scale-invariant spacing lemma (source lines 695-698), prop:icexponential (source lines 814-817) and the Mecke equation prel:mecke (source lines 1845-1852).

Editorial note. This article carries no label-number dump and the retained counter values are left empty for the whole source, so the printed numbering of the card is the wrapper's default; the identity of the counted item is its author label lem:prob e1c.

Editorial scope. One original result of the pinned publisher article, counted once: the estimate of order one over the logarithm for the failure of the first coupling event (author label lem:prob e1c).

Allowed standard prerequisites (editorial).

- The first coupling event. The event E1 controls the total mass of the point process and fails when some level exceeds the logarithm of x.. It is the event whose complement is estimated.
- Moments of the limiting variable. The exponential moment of the limiting variable is bounded uniformly, hence the probability that it exceeds the iterated logarithm is of the order one over the logarithm.. It removes the region where the level exceeds the iterated logarithm.
- Exponential law of the truncated weighted sum. The truncated weighted sum is exponentially distributed with parameter equal to the level.. It bounds the tail probability appearing after the expectation is computed.

$$\mathbb{E}[e^{\alpha\Theta_\infty}] = \exp\left(\int_0^\infty \frac{e^{\alpha r(y)} - 1}{y}\,\mathrm{d}y\right). \tag{1}$$

Lemma 0.1 (Scale-invariant spacing lemma)**.** *The point process* $\{S_i : i \in \mathbb{Z}\}$ *is a Poisson process on the positive real line with intensity measure* $\mathrm{d}s/s$.

Proposition 0.2 (Arratia, [2]). *For any fixed $t > 0$, the random variable I_t follows an exponential distribution of parameter t, i.e. $\mathbb{P}[I_t > y] = e^{-ty}$ for all $y > 0$.*

Proposition 0.3 (Mecke equation, [14, Th. 4.5]). *Let Π be a Poisson process on S with a σ-finite mean measure μ, and let $f : \Omega_S \times S^k \to [0, \infty)$ be measurable. Then we have*

$$\mathbb{E} \sum_{\substack{X_1,\dots,X_k \\ \text{all distinct}}} f(\Pi \setminus \{X_1, \dots, X_k\}; X_1, \dots, X_k) = \int_S \cdots \int_S \mathbb{E}[f(\Pi; x_1, \dots, x_k)] \, \mathrm{d}\mu(x_1) \cdots \mathrm{d}\mu(x_k).$$

Lemma 0.4. *We have $\mathbb{P}[\mathcal{E}_1^c] \ll 1/\log x$ for $x \geq 2$.*

Proof. We may assume that x is large enough. We have $\mathbb{E}[e^{R_x}] \leq \mathbb{E}[e^{\Theta_\infty}] \ll 1$ by (1). Therefore, the event $R_x > \log_2 x$ occurs with probability $\ll 1/\log x$. Using Lemma 0.1, we find that the probability that there exists $i \in \mathbb{Z}$ such that $|S_i - \log x| \leq 4r_0$ is $O(1/\log x)$. In conclusion,

$$\mathbb{P}[\mathcal{E}_1^c] \leq \sum_{k \in \{0,1\}} \mathbb{P}\Big[4r_0 < |S_k - \log x| \leq R_x + 2r_0, \ R_x \leq \log_2 x\Big] + O\big(1/\log x\big).$$

Note that if there exists $k \in \mathbb{Z}$ with $W_k > (\log_2 x)^{-10}$ and $|S_k - \log x| \leq \log_2 x + 2r_0$, then $I_{(\log_2 x)^{-10}} \geq \log x - \log_2 x - 2r_0$. So, using Proposition 0.2, we find

$$\mathbb{P}\Big[\exists k : |S_k - \log x| \leq \log_2 x + 2r_0, \ W_k > (\log_2 x)^{-10}\Big] \leq \mathbb{P}\big[I_{(\log_2 x)^{-10}} > \log x - \log_2 x - 2r_0\big] = \exp\Big(-\frac{\log x - \log_2 x - 2r_0}{(\log_2 x)^{10}}\Big)$$

for x large enough. Consequently,

$$\mathbb{P}[\mathcal{E}_1^c] \leq \sum_{k \in \{0,1\}} \mathbb{P}\Big[4r_0 < |S_k - \log x| \leq R_x + 2r_0, \ R_x \leq \log_2 x, \ W_k \leq (\log_2 x)^{-10}\Big] + O\big(1/\log x\big).$$

Let m_0 be the largest integer such that $2^{m_0 - 1} r_0 \leq \log_2 x$. Hence, if $2r_0 < R_x \leq \log_2 x$, then there exists a unique integer $m \in [2, m_0]$ such that $2^{m-1} r_0 < R_x \leq 2^m r_0$, in which case the condition $|S_k - \log x| \leq R_x + 2r_0$ implies that $|S_k - \log x| \leq 2^m r_0 + 2r_0 \leq 2^{m+1} r_0$. We conclude that

$$\mathbb{P}[\mathcal{E}_1^c] \leq \sum_{\substack{k \in \{0,1\} \\ 2 \leq m \leq m_0}} \sum \mathbb{P}\Big[R > 2^{m-1} r_0, \ |S_k - \log x| \leq 2^{m+1} r_0, \ W_k \leq (\log_2 x)^{-10}\Big] + O\big(1/\log x\big).$$

Note that $\mathbb{1}_{R_x > 2^{m-1} r_0} \leq (4^{1-m}/r_0^2) \mathbb{1}_{R_x > 2r_0} R_x^2$. In addition, if $R_x > 2r_0$, then we have $\sum_{i \geq 2} r(Y_i)^2 \leq r_0 R_x < R_x^2/2$, whence $R_x^2 \leq 4 \sum_{i > j \geq 2} r(Y_i) r(Y_j)$. We conclude that

$$\mathbb{P}[\mathcal{E}_1^c] \leq \sum_{\substack{k \in \{0,1\} \\ 2 \leq m \leq m_0}} \sum \frac{4^{2-m}}{r_0^2} \mathbb{E}\Big[\sum_{i,j:\ i > j > k} \sum r(Y_i) r(Y_j) \cdot \mathbb{1}_{|S_k - \log x| \leq 2^{m+1} r_0} \cdot \mathbb{1}_{W_k \leq (\log_2 x)^{-10}} \Big] + O\big(1/\log x\big),$$

Therefore, to complete the proof of the lemma, it is enough to show that

$$E(z) \coloneqq \mathbb{E}\Big[\sum_{i,j,k\in\mathbb{Z}:\ i>j>k} \sum\sum\sum r(Y_i)r(Y_j)\cdot \mathbb{1}_{|S_k-\log x|\le z}\cdot \mathbb{1}_{W_k\le(\log_2 x)^{-10}}\Big] \ll \frac{z}{\log x} \quad (2)$$

uniformly for $z \in [0, 4\log_2 x]$.

For the rest of the proof, we fix $z \in [0, 4\log_2 x]$. Given $t'' > t' > t > 0$, let

$$I_{t,t',t''} \coloneqq \sum_{\substack{(W,Y)\in\mathscr{R}\\ W\in\mathbb{R}_{>t}\setminus\{t',t''\}}} Y.$$

Since the W_i's are almost surely distinct, we have

$$E(z) \le \mathbb{E}\Big[\sum\sum\sum r(Y')r(Y'')\cdot \mathbb{1}_{|Y+Y'+Y''+I_{W,W',W''}-\log x|\le z}\cdot \mathbb{1}_{W\le(\log_2 x)^{-10}}\Big].$$

with the triple sum being over points $(W,Y),(W',Y'),(W'',Y'')\in\mathscr{R}$ satisfying $W''>W'>W$. Hence, using the Mecke equation (Proposition 0.3) and the fact that $I_{t,t',t''}$ has the same distribution as I_t, we find

$$E(z) \le \int\cdots\int_{\substack{0<w<w'<w''\\ w\le(\log_2 x)^{-10}\\ y,y',y''\ge 0\\ y+y'+y''\le\log x+z}} r(y')r(y'')\mathbb{P}\Big[\big|I_w+y+y'+y''-\log x\big|\le z\Big]\frac{\mathrm{d}w\cdots\mathrm{d}y''}{e^{wy+w'y'+w''y''}},$$

where the integral is sixfold with variables w, w', w'', y, y', y''. Proposition 0.2 implies

$$\begin{aligned}\mathbb{P}\Big[\big|I_w+y+y'+y''-\log x\big|\le z\Big] &\le e^{-w(\log x-y-y'-y'')}(e^{wz}-e^{-wz})\\ &\ll wze^{-w(\log x-y-y'-y'')},\end{aligned}$$

since $w\le(\log_2 x)^{-10}$ and $z\le 4\log_2 x$. Consequently,

$$\begin{aligned}E(z) &\ll \int\cdots\int_{\substack{0<w<w'<w''\\ y,y',y''\ge 0\\ y+y'+y''\le\log x+z}} r(y')r(y'')zwe^{-w\log x-(w'-w)(y'+y'')-(w''-w')y''}\,\mathrm{d}w\cdots\mathrm{d}y''\\ &= \frac{z}{(\log x)^2}\iiiint_{\substack{t,y,y',y''\ge 0\\ y+y'+y''\le\log x+z}} \frac{r(y')r(y'')}{y''(y'+y'')}te^{-t}\,\mathrm{d}t\,\mathrm{d}y\,\mathrm{d}y'\,\mathrm{d}y'',\end{aligned}$$

where we made the change of variables $t = w\log x$. Since $\int_0^\infty r(u)/u\,\mathrm{d}u \ll 1$ and $\log x + z \ll \log x$, relation (2) follows. This completes the proof of the lemma. □

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